

CHANDRIL MALLICK

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PROFESSIONAL SUMMARY

Full-stack developer and AI enthusiast with a strong foundation in building scalable ML pipelines and web applications. Proven experience in developing RAG-based chatbots, Computer Vision systems, and deploying models using FastAPI and Docker. Passionate about bridging the gap between research and production engineering in HealthTech and Automation domains.

EDUCATION

B.Tech in Computer Science & Engineering

2023 – 2027

Brainware University, Kolkata

- **Key Coursework:** Data Structures & Algorithms, Database Management, Artificial Intelligence, Operating Systems.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript (ES6+), Java, Dart, SQL, C++
- **AI/ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, OpenCV, LangChain, RAG
- **Backend Development:** FastAPI, Node.js, Express, RESTful APIs
- **Frontend Mobile:** React.js, Tailwind CSS, Flutter
- **DevOps Tools:** Docker, Git/GitHub, Linux (Bash), Firebase, Supabase, SQLite, MySQL

EXPERIENCE

AI Intern — Samsung Innovation Campus

2025

- Engineered NLP pipelines for text classification tasks, improving model accuracy by 15% through advanced pre-processing.
- Deployed machine learning models as scalable REST APIs using FastAPI, enabling real-time inference for client applications.
- Collaborated on dataset curation and annotation strategies, ensuring high-quality input for supervised learning models.

KEY PROJECTS

Intelligent Campus Platform – University-Trained AI System

🔗 Code

- Built a production-ready **institution-specific RAG platform** enabling students to query syllabi, notices, and policies with source-grounded answers.
- Designed a full-stack system using **FastAPI, React, LangChain, and FAISS** supporting document verification and real-time academic support.
- Improved response faithfulness by limiting generation to retrieved institutional context, reducing hallucinations in academic queries.

Hallucination-Aware RAG System

🔗 Code

- Developed a **hallucination-detection pipeline** for LLM outputs using source attribution, semantic similarity, and faithfulness scoring.
- Implemented evaluation metrics to compare vanilla RAG vs hallucination-aware RAG, achieving significant improvement in answer reliability.
- Designed for high-stakes domains such as education, healthcare, and legal document QA.

WebScraper-RAG – Automated Knowledge Ingestion System

🔗 Code

- Built an end-to-end pipeline that scrapes web content, cleans and chunks data, and converts it into a **queryable RAG knowledge base**.
- Integrated intelligent document parsing and embedding storage using FAISS for low-latency semantic search.
- Enabled citation-backed responses for improved transparency and trust in LLM outputs.

Neural Foundations – Deep Learning From Scratch

🔗 Code

- Reconstructed core deep learning components including **forward/backpropagation, optimizers, and activations** without high-level frameworks.
- Implemented auto-differentiation logic to understand gradient flow and model optimization internals.
- Strengthened foundational understanding for custom model research and performance debugging.

ACHIEVEMENTS & CERTIFICATIONS

- **Top 25 Finalist:** HP Dreams Unlocked (Tech Track) – Recognized for innovative approach in Intelligent Campus Platform.
- **Top 1000 Innovator:** National AI Competition – Selected from over 10,000 participants nationwide.